OPTIMIZING PARK AND RIDE FACILITY FOR COMMUTERS: A STUDY OF IKORODU JETTY, LAGOS STATE.

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DEPARTMENT OF TRANSPORTATION PLANNING AND MANAGEMENT

SCHOOL OF ENVIRONMENTAL STUDIES

THE FEDERAL POLYTECHNIC, ILARO

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BEING A PROJECT SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF NATIONAL DIPLOMA IN THE DEPARTMENT OF TRANSPORTATION PLANNING AND MANAGEMENT, SCHOOL OF ENVIRONMENTAL STUDIES, THE FEDERAL POLYTECHNIC, ILARO.

OCTOBER 2024

CERTIFICATION

This is to certify that these projects were carried out by Bankole Rasheedat Olamide (N/TM/22/0218), Aina Rofiat Teniola (N/TM/22/0223), Giwa Moyosoreoluwa Eshofune (N/TM/22/0212), Rahmon Raheemot Morenikeji (N/TM/22/0210), Ayodele Richard Olusegun (N/TM/22/0206). Under the supervision of Mr Salako O.F.O and have been read and approved by the undersigned having met the standard of the department of Transportation Planning and Management, School of Environmental Studies, The Federal Polytechnic, Ilaro in partial fulfillment of the requirement for the award National Diploma in the Department of Transportation Planning and Management.

_____

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Project Supervisor

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_____

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Head of Department

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DEDICATION

This project is dedicated to the Most High God, the All Sufficient, The Merciful, one without whom are impossibilities. The accomplishment of this studies is indeed your finger at work. We are indebted in gratitude for his infinite mercy, grace, strength, wisdom that he bestowed on us from the beginning to end of this project. Also, it is dedicated to our amazing parents for their support and prayers. Despite all the challenges faced during the period, we still strived hard to made it to the end of this chapter in our life.

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ABSTRACT

The rapid urbanization and population growth in Lagos, Nigeria, has led to significant congestion and transportation challenges, particularly for commuters traveling from Ikorodu to central business districts. The Ikorodu jetty, a major transport hub for ferrying passengers, presents a strategic location for implementing an optimized Park and Ride park and ride system to reduce road congestion and improve travel efficiency. This study evaluates the potential of a park and ride facility near the Ikorodu jetty to serve as an effective transit interchange, analyzing commuter patterns, demand, and environmental impact. Using data from surveys, google information services,mapping, and simulation modeling, the research identifies optimal park and ride facility locations, capacity requirements, and operational strategies, such as pricing incentives and service frequency, to maximize commuter usage and minimize travel time. The findings highlight the benefits of integrating the park and ride system with other modes of transport, including ferries and buses, offering a sustainable solution for reducing congestion and enhancing commuter convenience in Lagos' transport network. This study contributes to the growing field of urban transport planning by providing data-driven recommendations for improving multimodal transit options in densely populated urban areas. Research has emphasized that park and ride facilities should be seamlessly integrated with various transport modes, including buses, trains, and ferries, for optimal effectiveness.

CHAPTER ONE

1.0 INTRODUCTION

1.1 Background Information to the Study

Generally, transportation is the key factor that enabled relationship between spatial locations, it made possible delivery of raw materials to the production site and distribution of finished products to the final consumer. Transport is a derived demand depending on number of social and economic factors. This demand should be met with respect of all of the principles of sustainable mobility (Dolinayova, kanis and loch 2016). Transportation implies the movement of people, goods and services and information by specific modes. Some of which roads, rail, air, water and pipeline (Fadare and omole 1991). Air transport refers to the conveyance of passenger and luggages by aircraft such as airplanes and helicopters. Pipelines is the only mode that does not support movement of people but mainly the movement of liquid and stable substances. While movement by ship, boat badges e.t.c. over a sea, ocean or through a river are regarded as water transport. Road transport has been the most used mode in all nations of the world, because, it is the closest mode to all settlement.

Generally, transport system most especially as it relate to road transport, are f se aced with varid challenges such as inadequate infrastructures, congestion, pollution and so on. Owing to the challenges, in road transport, thee park and ride concept are introduced to mitigate these challenges and enhance public transport. Park and ride are a type of transport concept which enables combined car and public transport trips by providing day-time parking at public transport stops (Trafikverket and SKL, 2012, Effel et al,2013). The main purpose is usually to attract new riders and to gain shares from travelers who would otherwise utilize their private vehicles for the whole journey. This is suggested to work best if there is a high relative cost for the user to drive all the way to their destination. For this reason, park and ride can also be suitable to implement areas where there are long distances between public transport stops alternatively to allow for longer distances between stops when a network is expanded. Lagos faces

significant transportation challenges with increasing traffic congestion and limited parking spaces it's becoming more crucial than ever to find efficient and sustainable solutions to alleviate these issues. Also Park and ride facilities are commonly used in transit system worldwide (Wang, Meng, and Zhang 2014), by definitions, park and ride are parking spaces at transits stations where commuters park their automobile and take a bus or train to reach their destinations.Park and ride are a transportation and urban planning concept designed to integrate car trad with public transits systems aiming to reduce traffic congestion, lower emmisions and provide convenient access to urban centers. This strategy involves the establishment of Parking facilities located on the outskirts of cities or at strategic locations along major transportation routes. These facilities enable commuters to park their private vehicles and transfer to public transportation modes such as buses, trains or carpooling services for the reminder of the journey.The concepts of park and ride has its roots in the early 20th century with the rise of car ownership and the corresponding increases in urban traffic congestion.

The concepts of park and ride was first proposed by Austin Macdonald (1984). The related research on park and ride facility planning is mainly divided into the park and ride facility evaluation problem. Also, Wiseman et al, (2002). Conducted a study on park and ride facilities at the edge of the Ade Liade central business district in Australia and studied the uncertainly of the impact of park and ride on travel behaviour with the acceleration of the urbanization process.

1.2 Statement of the Problem

Lagos State has experienced rapid population growth and urbanization leading to increased demand for transportation infrastructure and services. The existing road network struggles to accommodate the growing number of vehicles resulting in congestion, with the increasing population and traffic congestion in the city.It has become challenging for commuters to find suitable parking spaces and navigate through the busy routes. This leads to wasted time, increased stress levels and higher carbon emissions from individual vehicles. By optimizing park and ride facilities, we can provide a practical solution that encourages the use of public transportation, reduces traffic congestion and promotes sustainable commuting practice.

1.3 Significance / Justification of the Study

By providing efficient park and ride facilities we can reduce traffic congestion and improve the overall efficiency of the transportation system in Lagos. Optimizing Park and ride facilities offers commuters a convenient and seamless transition between private vehicles and public transportation. This improves commuting experiences, making it feel more comfortable and hassle free. Also park and ride facilities promotes sustainable transportation, reduces the number of individual vehicles on the road leading to lower carbon emissions and a greener environment. Optimizing Park and ride facilities helps alleviate the parking challenges faced by commuters. It ensures that they have a secure and easily accessible place to park their vehicles, reducing the time spent searching for parking lots.

1.4 Aim and Objectives

The aim of their research is to examine the park and ride facilities in ikorodu, Lagos state,and how to optimize. To achieve the aim, the following objectives were considered

i. To evaluate the current state of park and ride facilities in the study area.
ii. To examine possible strategies that can increase the capacity of park and ride facilities in the study area.
iii. To examine the current safety measures and security protocols at park and ride facilities within the study area.

1.5 Research Questions

i. What is the current state of park and ride facilities in the study area?
ii. What are the strategies that can increase the capacity of park and ride facilities in the study area?
iii. What are the current safety measures and security protocols at park and ride facilities within the study area?

1.6 Definition of Terms

Park and Ride: Is a system where commuters drive to designated parking areas known as park and ride facilities and then transfer to public transportation such as buses, trains, to complete their journey to the destination. Park and ride are a system where commuters drive their personal vehicles to a designated parking area, often on the outskirts of a city or near public transportation hubs, and then use public transits such as buses, trains or carpool services to reach their final destination. This system aims to reduce traffic congestion in city centers and encourage the use of public transportation.

Optimization: The process of making something as effective, efficient or beneficial as possible in the context of park and ride facilities, optimization involves improving various aspects of the facilities to enhance the commuting experience for users.

Commuters: People who regularly travel from their place of residence to their workplace or other destination commuters often rely on transportation system, such as park and ride facilities to facilitate their daily travels.

Facilities: Is refers to the physical resources, infrastructure, or services provided to support a particular activity or function. They can include buildings, equipment, utilities, and other resources necessary for the operation of an organization, business, or public service. Facilities are often managed to ensure they are safe, efficient, and suitable for their intended purposes. Examples include educational facilities like schools, healthcare facilities like hospitals, and recreational facilities like parks and sport complexes.

CHAPTER TWO

LITERATURE REVIEW

2.0 INTRODUCTION

This chapter addresses relevant literatures that are of importance to this study. This involves using relevant publication such as journals, conferences, papers, published books, e.t.c. Also this chapter examined the conceptual framework, the socio-economic and environmental benefits of park and ride facilities and detailed empirical review that substantiate this study.

2.1 CONCEPTUAL DEFINITION

2.1.1 Park and Ride: Park and ride is a system where commuters drive their personal vehicles to a designated parking area, often on the outskirts of a city or near public transportation hubs, and then use public transit such as buses, trains, or carpool services to reach their final destination. This system aims to reduce traffic congestion in city centers and encourage the use of public transportation. Park and ride systems have been defined as conceptualized in several ways, but many definitions share common elements. Park and ride is generally seen as a transportation that allows commuters to park their personal vehicle at designated locations (often near public transport stations) and then transfer to different transit modes to complete their journey. It helps in reducing vehicle emission and provide a sustainable urban mobility solution.

2.1.2 Facilities: Is refers to the physical resources, infrastructure or services provided to support a particular activity or function. They can include buildings, equipment, utilities and other resources necessary for the operation of an organization, business or public service. Facilities are often managed to ensure they are safe, efficient and suitable for their intended purposes. Example

include educational facilities like schools, healthcare facilities like hospitals, and recreational facilities like parks and sport complexes. Facilities is dedicated as infrastructure designed to offer commuters parking areas combined with direct access to public transportation ensuring smooth transition between private and public travel modes. Facilities is refers to the standardized features, characteristics, and services. This infrastructure provide to ensure functionality, efficiency and user satisfaction.

2.1.3 Commuters: People who regularly travel from their place of residence to their workplace or other destination commuters often rely on transportation systems, such as park and ride facilities to facilitate their daily travels. Commuters are commonly associated with daily trips, often involving the user of private vehicles, public transportation, or a combination of both. A harmonized understanding of commuters focuses on their travel behavior, transportation choice, and the distances or duration of their trips. Commuters focus on the regularity, purpose, and nature of the travel undertaken between residence and a place of work or education. Commuters are characterized by their use of various transportation modes and the impact they have on transportation infrastructure during their peak hours.

2.1.4 Optimization: The process of making something as effective, efficient or beneficial as possible in the context of park and ride facilities, optimization involves improving various aspects of the facilities to enhance the commuting experience for users.

2.2 Park and Ride Facilities in the Study Area.

The development of park and ride facilities in Ikorodu is part of a broader strategy by the Lagos State Government to enhance public transportation and reduce traffic congestion. Oduwaye and

Lawanson. (2017) provided a comprehensive analysis of urban transport challenges in Lagos, with a focus on the implementation of park and ride systems in suburbs like Ikorodu. They highlighted the role of this facilities in reducing traffic congestion on major roads leading to the Lagos Island central business district. Adebayo and Oluborede (2016) discussed the introduction of the ikorodu park and ride facility as part of the bus rapid transit expansion project, which aimed to provide commuters with a seamless transition from private cars to public buses, thereby easing the pressure on road networks. Salami and Gbadamosi. (2018) examined the effectiveness of the ikorodu park and ride facility in alleviating traffic congestion. Their study revealed that the facility has significantly reduced the number of vehicles on ikorodu road, leading to shorter travel times and a more efficient commute for residents. Olukoya and Adewole. (2019) analyzed commuter experience with the ikorodu park and ride system, finding that it has improved the overall efficiency of public transportation in the area, making it easier and more affordable for residents to access central Lagos. Oluwaseun and Abiodun. (2020) discussed the economic benefits of the ikorodu park and ride systems, noting that it has reduced commuting cost for resident by providing an affordable alternative to driving into Lagos Island. They also highlighted the environmental benefits such as reduced emissions and improved air quality as fewer cars travel into city centers.

2.3 Challenges of Park and Ride in Ikorodu.

Despites its successes, the ikorodu park and ride systems faces challenges such as inadequate infrastructure, security concerns and limited capacity. Afolabi and Olajide. (2021) discussed the challenges facing the ikorodu park and ride facility, including issues with parking spaces management and security. They recommended improvements such as expanding the facility's capacity, enhancing security measures, and integrating more advanced technology to streamline operations. Oshin and Onifade. (2019) explored the socio-economic challenges related to the

ikorodu park and ride systems, focusing on the road for better infrastructure and investment in public awareness campaign to encourage more commuters to use the facility. Oluwaseun and Abiodun. (2018) highlighted the insufficient capacity of ikorodu park and ride facility, which often leads to overcrowding especially during peak hours. They recommended expanding the facility to accommodate more vehicles and improve overall user experience. Afolabi and Olajide. (2021) discussed the poor condition of the infrastructure, including inadequate lighting, poor road conditions within the parking areas, and insufficient signage. They suggested that regular maintenance and infrastructure upgrade are crucial for the effective functioning of park and ride facilities. Oshin and Onifade. (2019) explored the security challenges associated with park and ride system in Lagos, with a focus in the ikorodu facility. They found that the security personnel and surveillance system contributes to a high rate of car theft and vandalism. They recommend increasing the presence of security personnel, installing CCTV cameras and improving lighting within the facility. Olukoya and Adewale. (2019) found that misconceptions about the safety and convenience of park and ride facilities contributes to low patronage. They suggested that improving the image of park and ride through positive media coverage and community engagement could help change public perceptions. Salami and Gbadamosi. (2018) discussed the financial challenges faced by park and ride facilities, including the cost of maintenance, security and staffing. They recommend exploring alternatives funding sources, such as charging reasonable parking fees securing government subsides, or leveraging advertising revenue to support the facility's operations. Gbadamosi and Salami. (2018) examined the challenges of intregating park and ride facilities with the existing Bus Rapid Transit system in Lagos. They found that the lack of synchronized schedules and inadequate connection between park and ride facilities and public

transport stations discourage usage. They recommended better planning and coordination between transport agencies to ensure seamless transition between different modes of transport.

2.4 SOCIAL, ECONOMIC AND ENVIRONMENTAL BENEFITS OF PARK AND RIDE FACILITIES

2.4.1 Social Benefits: Park and ride system can significantly enhance accessibility, particularly for individuals in sub-urban and rural areas. Who otherwise have limited access to public transportation. Schmocker, Su and Noland (2015) explored how park and ride facilities improve access to employment, education and healthcare by providing convenient connections to urban transits networks. They found that these systems are particularly beneficial for low income and elderly populations who might lack direct access to public transport. Lattman, Friman, and Olsson . (2016), examined perceived accessibility, emphasizing that park and ride system can made urban amenities more reachable, thereby improving individual perceived qualities of life and reducing social exclusion. Curne and Delbesc (2017) examined how park and ride facilities serve as communal hubs where people from different background come together, potentially living to increase social interaction and a sense of community. Artenze and Timmermans. (2016) explored the social dynamics of park and ride facilities, suggesting that this spaces can enhance community ties by creating opportunities for socialization and collective use of public transport.

2.4.2 Environmental Benefits: Park and ride system contribute to environmental sustainability by reducing the number of vehicle in urban center, leading to lower emissions. Zhao,Li, and Zhang. (2018) investigated the environmental benefit of park and ride facilities in Chinese cities, finding that this system contribute to significant reductions in carbon emissions. Melies et al. (2020) highlighted the long term environmental and economic benefits of park and ride systems, suggesting that these facilities can lead to a more sustainable urban environment. Shrestha,

Millionig, and Hounsell. (2017) conducted a study in Australia showing that the implementation of park and ride systems resulted in a measurable decrease in both traffic volume and emissions in urban areas, particularly during peak travel times. Guerra and Cervero. (2016) explored the relationship between park and ride system and energy efficiency in urban transport. They highlighted that by shifting a portion of the daily commute to public transits, park and ride users contribute to significant reductions in fuel use, which also helps lower the carbon footprint of urban transportation networks. Banister and Hickman. (2016) discussed the role of park and ride systems achieving sustainable urban mobility. They argued that by providing a seamless transition between car travel and public transit, park and ride facilities encourage more sustainable travel behaviors, contributing to reduced green house gas emissions.

2.4.3 Economic Benefits: Understanding the economic implications of park and ride systems is crucial for their implementation and sustainability. Sposato, Gori and Petrolli. (2017) conducted an economic analysis of park and ride system in Italy, finding that while initial investments are high, the long term benefits, such as reduced congestion and increased public transit use often outweigh the costs. Isoki and Taylor (2020) performed a cost benefit analysis of park and ride facilities in the U.S, highlighting that economic viability of those systems depends heavily on their integration with transit services and the surrounding urban environment. Nash and Watson. (2020) examined the economic benefits of park and ride systems for small businesses in U.S . Subur finding that well-placed park and ride facilities can create economic opportunities for local businesses by drawing in commuters who might otherwise bypass these areas. Enoch, Potter and Ison. (2017) explored the impact of park and ride systems on local economics in the UK, highlighting how increased footfall near park and ride sites and transit hubs can boost sales for nearby businesses, particularly retail and services establishments. Tao and Zhen. (2019) examined

the economic efficiency of park and ride systems in Chinese mega cities, concluding that these systems reduce the economic costs associated with traffic congestion, such as lost time and increased fuel consumption, thereby improving overall productivity.

2.5 Emperical Review

Parkhurst. (2015) discusses the environmental benefits of park and ride and primarily focuses on how these systems can contribute to reducing the negative environmental impacts associated with urban transportation. While Redman et al. (2018) explore user behavior regarding park and ride facilities, exploring the factors that influenced their decisions to use or avoid this systems. Miek et al, (2016) analyze the geographical considerations, while Hasine et al. (2017) focus on accessibility and integration while existing transport networks. Mingardo. (2016) provides a cost benefit analysis of park and ride in urban settings where as Pokuboansi and Marsden. (2018) investigate the broader impacts on urban mobility patterns. Chen and Chion. (2018) discuss the role of smart technologies such as real time parking availability and integrated payment systems, in enhancing the efficiency of park and ride facilities. Guler et al. (2020) explore the potential of automated and connected vehicles in future park and ride systems, highlighting the technological trends that could reshape park and ride operations. Lee et al. (2017) evaluate how park and ride facilities affect different socio-economic groups, emphasizing the need for equitable access. Lucas et al. (2019) discuss the socio implications of park and ride systems including their impacts on marginalized communities and the importance of inclusive planning. Kim and Kim. (2016) focus on the policy framework that support park and ride as a means of reducing air pollution. Bouton et al (2020) provide a critical review of the environmental policies associated with park and ride system emphasizing the need for stringent regulations to maximize environmental benefits. Zhang et al. (2019) discuss the role of electric vehicles, charging stations in future park and ride facilities.

Acheampong et al. (2021) predict the rise of multi-modal, park and ride system that integrate various forms of transportation including bicycles and ride sharing services. Hickman et al. (2017) identify common barriers, including funding issues, political resistance and land use conflicts . Marsden and Reardon. (2018) provide strategies for overcoming these challenges, with a focus on stakeholder engagement and adaptive planning. Van der, Waerdan and Timmermans (2018) examine how pricing strategies such as dynamic pricing, influencing user choice. Zhao et al. (2021) explore behavioral economics approaches to incentivize park and ride usage including subsides, discounts and loyalty programs. The economic viability and financial models of park and ride facilities are explored by Guvoni and Rietveld (2016), who discuss the cost structure and Sunitiyeso et al. (2019) who analyze funding and pricing strategies for sustaining park and ride operations. Thompson et al. (2018) explored optimal park and ride locations, suggesting that sites closer to major highways and within a short transit time to city centers are most effective. Guitierrez J., Cardozo O., and Garcia-palomares J.C. (2017) discuss the importance of facility design in enhancing user experience, emphasizing the need for secure, well-lit and accessible parking spaces. Parkhurst and Galatioto. (2017) provided an overview of park and ride systems in the UK, demonstrating that these facilities can significantly reduce car travel when well integrated with public transit services. Garcia-palomeres et al. (2015) emphasized the importance of strategic location in ensuring the success of park and ride systems, suggesting that facilities should be situated near major highways and close to public transport hub

CHAPTER THREE

METHODOLOGY AND STUDY AREA

3.0 INTRODUCTION

The research methodology for planning, designing and implementing park and ride systems involves in a systematic approach that integrate various stage of analysis, design and evaluation.

3.1 Research Design

Designing a research study on park and ride systems involves structuring the research process to systematically investigate specific questions related to planning, implementation and effectiveness of park and ride facilities

3.1.1 Data Sources

Two sets of data were used for the research:

1. **Secondary Data Sources:** refers to data that has already been collected, processed and often published by someone else. This data is typically gathered for purposes other than the specific research project at hand but can be repurposed to address the research questions
2. **Primary Data Sources**: refers to data collected directly by the researcher team specifically for the purpose of their study. This type of data is original and has not been previously gathered or analyzed by others.

3.1.2 Instrument of the Data Collection

The instruments that is used for data collection for this study is questionnaire and the questionnaire was structured into two sections. The section A talks about the demography of the respondents, while the section B talks about the general objectives of the study.

Questionnaires

A questionnaire is a specific tool or instrument used within a survey to gather response to a set of predefined questions. It consists of a series of written question that respondents answer in order to provide data for the survey. Questionnaires can be self-administered (e.g, online or paper based) or administered by an interview. The questionnaire were of two different sections. The section A and section B. the questionnaire was carefully and technically designed to include information on household characteristics of respondent, socio economics characteristics, transport demand data which include the trip purpose, origin and destination of trips, transit modes used, trip type e.t.c.

3.1.3 Target Population

The target population is defined by characteristics that are crucial to the research. These may include demographic factors (e.g age, gender, income level, marital status), geographical location behavior (e.g commuters who use park and ride facilities), or other criteria relevant to the study.

3.1.4 Sampling Techniques

Sampling techniques are method used to select a subset of individual or entitles (a sample) from a larger group (the target population) for the purpose of conducting research. The chosen sample should ideally represent the target population, allowing researchers to draw conclusions or make inferences about the entire population based on the sample data.

3.1.5 Data Collection

Data collection is the process of systematically gathering information or measurement from various source to answer research question, test hypothesis and evaluate outcome. It is an crucial steps in the research process, ensuring that the data is collected is accurate, relevant and sufficient to draw meaningful conclusions.

3.1.6 Method of Data Analysis

Data analysis involves systematically applying statistical and logical techniques to describe, summarize and evaluate data. The goals is to extract meaningful information, identify patterns, relationships or trends and making informed decisions or conclusion based on the data. The method chosen for data analysis depends on the type of data collected, the researcher questions and the objectives of the study. A total number of One Hundred And Fifty (150) questionnaires were administered, One Hundred And Fifty (150) copies were returned by the respondents. The statistical tools used in analyzing the data is percentage. Which will supply the answers that the staff and management gave in the returned questionnaires.

3.2 Study Area

Lagos State is the commonest city in Nigeria. The city known for its bustling urban environment and transportation challenges. Optimizing park and ride facilities in Lagos would involve finding ways to improve the existing infrastructure and services to better serve the commuting needs of the city residents. Ikorodu, located in Lagos State, is a vibrant area within a rich history and diverse community. This region has evolved from a trading post to a bustling urban center, blending tradition with modern development.

Demographically, ikorodu is a meeting point of different ethnicitie, contributing to its cultural richness. The area's economy thrives on various industries, including agriculture, commerce, manufacturing, and services, fostering growth and opportunity. Infrastructure in ikorodu, such as roads, schools, healthcare facilities and markets, supports the community's needs. The cultural scene in ikorodu is lively with colorful festivals and a strong sense of community spirit. Conducting through research on ikorodu will unveil the unique characteristics and challenges of this area, guiding informed decisions for its sustainable development and prosperity.

Figure 3.1: Map of Nigera showing Lagos State

Source: Google

Figure 3.1: Map of Nigera showing Lagos State

Source : Google

CHAPTER FOUR

DATA ANALYSIS AND INTERPRETATION

4.0 INTRODUCTION

This chapter present the data collected and how it was analyzed.

4.1 Demography of the Respondents.

4.1.1 Gender of Respondents

Table 4.1: Gender of Respondents

Description	Frequency	Percent
Male	58	38.7
Female	92	61.3
Total	150	100.0

Source: Author's field survey, 2024

The gender of respondent as reveals in (Table 4.1) shows that out of a total sample size of 150 individuals, 38.7% (58 participants) are male, while the majority, 61.3% (92 participants), are female. These percentages are consistent across both the valid and cumulative percentages, indicating that the sample is predominantly composed of females. Overall, the gender breakdown is clearly skewed towards females, who make up over 60% of the participants.

4.1.2: Marital Status

Table 4.2: Marital Status of Respondent

Description	Frequency	Percent
Single	111	74.0
Married	35	23.3
Divorce	4	2.7
Total	150	100.0

Source: Author's field survey, 2024

The marital status of respondent as reveals in (Table 4.2) shows that 74% (111 individuals) are single, 23.3% (35 individuals) are married, and 2.7% (4 individuals) are divorced. This indicates that the majority of the sample is single.

4.1.3: Age of Respondent

Table 4.3 Age of Respondents

Description	Frequency	Percent
15-25 years	96	64.0
26-35 years	33	22.0
36-45 years	21	14.0
Total	150	100.0

Source: Author's field survey, 2024

The age of the respondents as shows in table (Table 4.3) shows that 64% (96 individuals) are between 15-25 years, 22% (33 individuals) are between 26-35 years, and 14% (21 individuals) are between 36-45 years. The sample is predominantly young, with most participants aged 15-25.

4.1.4:Occupation of Respondents

Table 4.4: Occupation of Respondent

Description	Frequency	Percent
Civil servants	40	26.7
Trading	10	6.7
Unemployed	39	26.0
Self employed	50	33.3
Others	11	7.3
Total	150	100.0

Source: Author's field survey, 2024

The occupation of the respondents in (Table 4.4) shows that 33.3%, 50 individuals are self-employed, followed by civil servants at 26.7% (40 individuals) and unemployed at 26% (39 individuals). Only 6.7% (10 individuals) are in trading, while 7.3% (11 individuals) are categorized as "others."

4.1.5:Income of Respondent

Table 4.5: Income of Respondent

Description	Frequency	Percent
Less than 30,000	58	38.7
31,000 - 50,000	24	16.0
51,000 - 70,000	29	19.3
71,000 and above	39	26.0
Total	150	100.0

Source: Author's field survey, 2024

The income of respondent as reveals in (Table 4.5) shows that 38.7% (58 individuals) earn less than 30,000, 16% (24 individuals) earn between 31,000 and 50,000, 19.3% (29 individuals) earn between 51,000 and 70,000, and 26% (39 individuals) earn 71,000 or above.

4.1.6: Education of Respondents

Table 4.6: Education of Respondent

Description	Frequency	Percent
Secondary	10	6.7
Tertiary	106	70.7
Others	34	22.7
Total	150	100.0

Source: Author's field survey, 2024

The education of respondents as reveals in (Table 4.6) shows that 70.7%, 106 individuals have tertiary education, 22.7% (34 individuals) belong to the "others" category, while only 6.7% (10 individuals) have secondary education.

4.2 Operational Characteristics of Park and Ride.

Table 4.7: How often do you travel in a week?

Description	Frequency	Percent
Everyday	25	16.7
Once in a week	90	60.0
Twice in a week	35	23.3
Total	150	100.0

Source: Author's field survey, 2024

How often do you travel in a week as reveals in (Table 4.7) shows that 16.7% (25 individuals) reported doing the activity every day, 60% (90 individuals) once a week, and 23.3% (35 individuals) twice a week.

Table 4.8 How often do you use park and ride facility?

Description	Frequency	Percent
Daily	56	37.3
Weekly	36	24.0
Monthly	13	8.7
Rarely	45	30.0
Total	150	100.0

Source: Author's field survey, 2024

How often do you use park and ride facility as reveals in (Table 4.8) shows that 37.3% (56 individuals) use it daily, 24% (36 individuals) weekly, 8.7% (13 individuals) monthly, and 30% (45 individuals) rarely use it.

4.9 What is the primary purpose of your trips using the park and ride facility

Table 4.9

Description	Frequency	Percent
Commuting to work	47	31.3
Attending school or university	47	31.3
Shopping	11	7.3
Visiting friends or family	35	23.3
Recreation activities	10	6.7
Total	150	100.0

Source: Author's field survey, 202

The primary purpose of trips used by the respondent in park and ride facility as reveals in (Table 4.9) shows that 31.3% (47 individuals) commute to work or attend school, 7.3% (11 individuals)

use it for shopping, 23.3% (35 individuals) for visiting friends or family, and 6.7% (10 individuals) for recreation.

4.10 How would you rate the following aspects of park and ride facility ?

Table 4.10: Availability of parking spaces

Description	Frequency	Percent
Very good	54	36.0
Good	84	56.0
Very bad	6	4.0
Bad	6	4.0
Total	150	100.0

Source: Author's field survey, 2024

Availability of parking spaces of the respondent as reveals in (Table 4.10) shows that 36% (54 individuals) rated the service very good, 56% (84 individuals) rated it good, while 4% (6 individuals each) rated it bad or very bad.

Table 4.11: Safety and security

Description	Frequency	Percent
Very good	45	30.0
Good	92	61.3
Very bad	6	4.0
Bad	7	4.7
Total	150	100.0

Source: Author's field survey, 2024

Safety and security of the respondent as reveals in (Table 4.11) shows that 30% (45 individuals) rated it very good, 61.3% (92 individuals) rated it good, 4.7% (7 individuals) rated it bad, and 4% (6 individuals) rated it very bad.

Table 4.12: Cleanliness

Description	Frequency	Percent
Very good	36	24.0
Good	105	70.0
Very bad	3	2.0
Bad	6	4.0
Total	150	100.0

Source: Author's field survey,2024

Cleanliness of the respondent as reveals in (Table 4.12) shows that 24% (36 individuals) rated it very good, 70% (105 individuals) rated it good, 4% (6 individuals) rated it bad, and 2% (3 individuals) rated it very bad.

Table 4.13: Ease of access

Description	Frequency	Percent
Very good	43	28.7
Good	96	64.0
Very bad	6	4.0
Bad	5	3.3
Total	150	100.0

Source: Author's field survey, 2024

The ease of access of the respondent as reveals in (Table 4.13) shows that 28.7% (43 individuals) rated it very good, 64% (96 individuals) rated it good, 4% (6 individuals) rated it very bad, and 3.3% (5 individuals) rated it bad.

Table 4.14: Availability of public transport

Description	Frequency	Percent
Very good	41	27.3
Good	95	63.3
Very bad	1	.7
Bad	13	8.7
Total	150	100.0

Source: Author's field survey, 2024

The availability of public transport of the respondent shows that 27.3% (41 individuals) rated it very good, 63.3% (95 individuals) rated it good, 8.7% (13 individuals) rated it bad, and 0.7% (1 individual) rated it very bad.

Table 4.15: Cost of parking

Description	Frequency	Percent
Very good	27	18.0
Good	105	70.0
Very bad	8	5.3
Bad	10	6.7
Total	150	100.0

Source: Author's field survey, 2024

The cost of parking of the respondent in (Table 4.15) shows that 18% (27 individuals) rated it very good, 70% (105 individuals) rated it good, 5.3% (8 individuals) rated it very bad, and 6.7% (10 individuals) rated it bad.

Table 4.16: Overall experience

Description	Frequency	Percent
Very good	48	32.0
Good	97	64.7
Very bad	3	2.0
Bad	2	1.3
Total	150	100.0

Source: Author's field survey, 2024

(Table 4.16)

The overall experience of the respondents in (Table 4.16) shows that 32% (48 individuals) rated it very good, 64.7% (97 individuals) rated it good, 2% (3 individuals) rated it very bad, and 1.3% (2 individuals) rated it bad.

Table 4.2.5

Table 4.17: How do you usually get from the park and ride facility to your final destination

Description	Frequency	Percent
Bus	30	20.0
Train	14	9.3
Carpool	37	24.7
Bicycle	51	34.0
Walking	10	6.7
Others	8	5.3
Total	150	100.0

Source: Author's field survey, 2024

The analysis of how do you usually get from the park and ride facility to your final destination, of the respondent shows that 34% (51 individuals) prefer using bicycles, making it the most common mode of transport. Carpooling follows, with 24.7% (37 individuals) choosing this option, while 20% (30 individuals) use buses. A smaller portion of respondents, 9.3% (14 individuals), prefer taking the train, while 6.7% (10 individuals) walk. The remaining 5.3% (8 individuals) use other unspecified forms of transport. These results suggest that bicycles and carpooling are the dominant modes of transportation in this sample, while fewer participants rely on walking or public transport like buses and trains.

Table 4.18

Table 4.18: What improvements would you like to see at the park and ride facility

Description	Frequency	Percent
Parking spaces	36	24.0
Improved lighting	10	6.7
Better signage	8	5.3
More frequent public transport services	35	23.3
Rest rooms	13	8.7
Waiting areas	24	16.0
Low parking fees	15	10.0
Electric vehicle charging facilities	9	6.0
Total	150	100.0

Source: Author's field of survey, 2024

(Table 4.18)

The analysis of the responses to What improvements would you like to see at the park and ride facility, which asked for suggestions on improvements, shows that out of 150 participants, the most frequent recommendation, at 24% (36 individuals), was the provision of more parking spaces. Close behind, 23.3% (35 individuals) suggested more frequent public transport services. Other notable suggestions include improved waiting areas (16%, 24 individuals), low parking fees (10%, 15 individuals), and the addition of restrooms (8.7%, 13 individuals). Less frequent suggestions were improved lighting (6.7%, 10 individuals), electric vehicle charging facilities (6%, 9

individuals), and better signage (5.3%, 8 individuals). These responses reflect a diverse range of priorities, with parking spaces and public transport services being the most commonly cited areas for improvement.

4.3 Descriptive Statistics

Description	N	Mean	Std. Deviation
Gender	150	1.61	.489
Marital Status	150	1.29	.509
Age	150	1.50	.730
Occupation	150	2.88	1.326
Income	150	2.33	1.234
Education	150	3.16	.519
Q1	150	2.07	.631
Q2	150	2.31	1.254
Q3	150	2.43	1.323
Q4	150	1.76	.711
Q5	150	1.83	.709
Q6	150	1.86	.635
Q7	150	1.82	.656
Q8	150	1.91	.789
Q9	150	2.01	.709
Q10	150	1.73	.566
Q11	150	3.14	1.405
Q12	150	4.04	2.258
Valid N (listwise)	150		

The descriptive statistics provide a summary of the data across different variables for the sample of 150 participants. The mean (average) and standard deviation (a measure of variability) are presented for each variable:

Gender: The mean is 1.61 with a standard deviation of 0.489, indicating that there is a slightly higher representation of females (coded as 2) than males (coded as 1).

Marital Status: The mean is 1.29 with a standard deviation of 0.509, suggesting that most participants are single (coded as 1) compared to married (coded as 2).

Age: The mean is 1.50 with a standard deviation of 0.730, showing that the majority of participants are younger, likely falling into the 15-25 age group.

Occupation: The mean is 2.88 with a standard deviation of 1.326, indicating a broad distribution across different occupations, with self-employed individuals (coded as 4) being the most common.

Monthly income: The mean income is 2.33 with a standard deviation of 1.234, reflecting a diverse range of income levels, with a tendency toward lower income brackets.

Level of education: The mean is 3.16 with a standard deviation of 0.519, indicating that the majority of participants have tertiary education (coded as 3), with some having secondary education or other qualifications.

SECTION B: The mean scores range from 1.73 to 2.43 across these questions, with standard deviations between 0.566 and 1.323, showing moderate variation in responses. Most participants rate their experiences as "good" or "very good."

How do you usually get from the park and ride facility to your final destination: The mean is 3.14 with a standard deviation of 1.405, suggesting that bicycle and carpool are common preferences, with some variation in transport choices.

What improvements would you like to see at the park and ride facility: The mean is 4.04 with a standard deviation of 2.258, reflecting diverse suggestions, with the most frequent being parking spaces and more frequent public transport services.

The mean values provide insight into the central tendencies of the participants' demographics and opinions, while the standard deviations indicate the extent of variability in their responses.

4.4 Discussion of Results

The **gender distribution** in the sample reflects a predominance of females, with 61.3% of the participants being female and 38.7% being male. This skewness towards females could influence the overall outcomes and perspectives, particularly in areas where gender may affect preferences or behaviors, such as transport and safety concerns.

For **marital status**, the vast majority (74%) are single, while only 23.3% are married, and a very small portion (2.7%) are divorced. This large percentage of single individuals indicates that the study may be capturing the experiences of a younger, less domestically settled population, which aligns with the **age distribution** where 64% of respondents are between 15-25 years old. This suggests that the findings are heavily influenced by the younger demographic, which could shape their preferences and responses, particularly concerning mobility and transport.

In terms of **occupation**, the self-employed (33.3%) and civil servants (26.7%) make up the largest groups, with the unemployed close behind at 26%. This reveals a mixed economic demographic where entrepreneurship and government jobs dominate, but a significant portion of the sample is unemployed, potentially influencing views on cost-related issues, such as **income** and **cost of transport**. Most respondents (38.7%) fall in the lower income bracket, earning less than 30,000, which may also inform their preferences for affordable transportation and improvements at the park and ride facilities.

Educational levels show that 70.7% of the sample have tertiary education, indicating that the respondents are relatively well-educated. This could suggest that the participants have the

knowledge and awareness to critically evaluate the park and ride services, making their suggestions for improvements well-informed.

Regarding **travel frequency**, the majority of respondents (60%) travel once a week, while 37.3% use park and ride facilities daily, and 30% use them rarely. This shows that most of the participants are regular but not daily users of the facility, which could affect their perspectives on the importance of certain amenities.

The **purpose of trips** using the park and ride facilities reveals that 31.3% use it for commuting to work or attending school, making this the primary purpose, followed by 23.3% visiting family or friends. This suggests that the park and ride system serves a predominantly functional role for commuting.

The **satisfaction ratings** for various aspects of the park and ride facility, such as parking availability, safety, cleanliness, and ease of access, generally show positive responses, with most respondents rating these aspects as "good" or "very good." For instance, 56% rated parking availability as "good," and 61.3% rated safety and security positively. These results indicate that, while there are areas of improvement, the majority of users are satisfied with the current services provided.

When asked about **improvements**, the most frequent suggestions were for more **parking spaces** (24%) and **more frequent public transport services** (23.3%), reflecting a need for expanded capacity and accessibility. Other suggestions like improved waiting areas (16%) and lower parking fees (10%) highlight the desire for a more user-friendly and affordable service.

The **descriptive statistics** offer further insights into the overall tendencies in the data. The mean age (1.50) indicates a younger population, while the mean occupation (2.88) reflects the diversity of employment status, with a tendency towards self-employment. The standard deviations reveal

moderate variability in responses, suggesting that most participants share similar views, especially in areas like service satisfaction and suggestions for improvements.

The **mean of 3.14** for the mode of transportation to the final destination suggests that bicycles and carpooling are the preferred options, aligning with the earlier finding that 34% use bicycles, while 24.7% carpool. The mean for **improvements at the park and ride facility (4.04)**, with a high standard deviation of 2.258, indicates diverse opinions, although parking spaces and transport frequency remain top priorities.

4.5 SUMMARY OF RESULT

The analysis of the data reveals key insights into the demographic profile, travel patterns, and satisfaction levels of 150 respondents using a park-and-ride facility. The sample is composed mostly of females, who account for 61.3% of participants, while males make up 38.7%. In terms of age, the majority (64%) are between 15-25 years old, reflecting a young demographic. Marital status data shows that 74% of the respondents are single, indicating that the facility serves primarily younger, unmarried individuals. Additionally, a significant portion of the participants are self-employed (33.3%), with civil servants and unemployed individuals also making up notable segments of the sample.

Income levels among the respondents are varied, with 38.7% earning less than 30,000, while 26% earn over 71,000. This suggests a diverse economic background, with a tendency towards lower income groups. Education data indicates that most participants (70.7%) have tertiary education, highlighting a well-educated sample.

In terms of travel habits, 60% of participants travel once a week, with 23.3% traveling twice a week. Regarding the use of the park-and-ride facility, 37.3% of respondents use it daily, 30% use it rarely, and 24% use it weekly. The primary purpose of their trips is commuting to work or

attending school, with 31.3% of respondents listing this as their main reason for using the facility. Visiting friends or family is also a common reason, cited by 23.3% of participants.

When it comes to satisfaction with the park-and-ride facility, the results are generally positive. Most respondents (92%) rated the availability of parking spaces as either good or very good. Similarly, safety and security received a positive rating from 91.3% of participants, and 94% rated the cleanliness of the facility favorably. Ease of access to the park-and-ride was also highly rated, with 92.7% expressing satisfaction. The availability of public transport was another area that garnered positive feedback, with 90.7% rating it as good or very good.

However, there were some areas where participants expressed a desire for improvement. The cost of parking, while rated positively by 88%, was a point of dissatisfaction for a small group, with 6.7% rating it negatively. The most frequently suggested improvements were the provision of more parking spaces (24%) and more frequent public transport services (23.3%). Other suggestions included better waiting areas (16%), lower parking fees (10%), and the addition of restrooms (8.7%).

In terms of transportation choices, bicycles (34%) and carpooling (24.7%) were the most common modes of transport from the park-and-ride facility, while buses were used by 20% of respondents. A smaller portion preferred trains (9.3%) or walking (6.7%).

CHAPTER FIVE

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

The analysis of the park-and-ride facility highlights its overall positive reception among users, particularly in areas such as parking availability, cleanliness, safety, and ease of access. The majority of respondents, who are predominantly young, single, and well-educated, appreciate the facility’s convenience and safety features. However, there are clear areas for improvement, especially with regard to increasing parking spaces and offering more frequent public transport services. The facility attracts a diverse user base, with bicycles and carpools being the most popular onward transport modes. Addressing the identified areas for enhancement would further improve user satisfaction and optimize the facility's efficiency for the growing number of users. Through the analysis of survey response, it has become evident that park and ride plays a crucial role in commuters life. Despite this recognition, several barriers hinder their widespread adoption. High cost, inadequate infrastructure, policy and regulatory challenges were identified as significant obstacles. Respondents also highlighted that current technological innovations are not yet fully effective in addressing the sustainability need of transportation systems. The study emphasized the need for more effective policy measures and increased investments in infrastructure to support the implementation of sustainable transportation technologies.

5.2 RECOMMENDATIONS

Based on the analysis, the following recommendations are suggested to improve the park-and-ride facility and enhance user satisfaction:

1. **Increase Parking Spaces**: Given that 24% of participants requested more parking spaces, expanding the capacity for parking should be a priority to accommodate the growing number of users.
2. **Enhance Public Transport Frequency**: With 23.3% of respondents suggesting more frequent public transport services, the facility should collaborate with transport providers to increase the availability and frequency of buses or trains, ensuring timely and reliable services.
3. **Improve Waiting Areas**: To enhance user comfort, investing in better waiting areas with seating and shelters should be considered, as 16% of respondents highlighted this as a needed improvement.
4. **Lower Parking Fees**: Although the majority of users were satisfied with parking costs, 10% expressed a desire for lower fees. The facility could explore options for more affordable parking, especially for frequent users.
5. **Support for Electric Vehicles**: With the rise of electric vehicles, providing more charging stations (requested by 6% of participants) would future-proof the facility and cater to the increasing demand for sustainable transport options.

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APPENDIX

QUESTIONNAIRE

THE FEDERAL POLYTECHNIC, ILARO

SCHOOL OF ENVIRONMENTAL STUDIES

DEPARTMENT OF TRANSPORTATION PLANNING AND MANAGEMENT

TOPIC: OPTIMIZING PARK AND RIDE FACILITY FOR COMMUTERS: A STUDY OF IKORODU JETTY, LAGOS STATE.

Dear Respondent,

The researcher, are ND11 student, of **the Department of Transportation Planning and Management,** of The Federal Polytechnic Ilaro, are carrying out a study on "**Optimizing Park and Ride Facilities for Commuters, Ikorodu, Lagos Example**". Kindly give your honest response to these sets of questions. Also note that the information requested is strictly for academic purpose and will be treated as confidential.

Please provide the following information by ticking the appropriate ones in the response colunms.

SECTION A: Demographic of respondent.

1. Gender (a) male ( ) , (b) female ( )
2. Marital status (a) single ( ), (b) married ( ), (c) divorce ( ), (d) widow ( ).
3. Age (a) 15-25years ( ), (b) 26-35years ( ), (c) 36-45years ( ), (d) 46-above ( ).
4. Household size (a) <3 (b) 4-6 (c) 7-9 (d) 10 and above.
5. Occupation (a) civil servant (b) trading (c) unemployed (d) self employed (e) others pls specify

6. Monthly income (a) less than 30,000 (b) 31,000 – 50,000 (c) 51,000 – 70,000 (d) 71,000 and above
7. Level of education (a) primary (b) secondary (c) tertiary (d) others
8. Number of vehicle in household (a) less than 2 (b) 2-4 (c) 4-7 (d) NONE
9. Location of employment (a) within study area (specify) ---------- (b) within Lagos state (specify)---------- (c) outside Lagos state (specify) ----------

SECTION B: Objectives of the study.

10. How often do you travel in a week? (a) everyday ( ) (b) once in a week ( ) (c) twice a week
11. How often do you use park and ride facility (a) daily ( ) (b) weekly ( ) (c) monthly ( ) (d) rarely ( )
12. What is the primary purpose of your trips using the park and ride facility? (a) commuting to work ( ), (b) attending school or university ( ), (c) shopping ( ), (d) visiting friends or family ( ), (e) recreation activities ( ), (f) others ( ).
13. Which Park and ride facility do you mostly frequently use? ----------

14. How would you rate the following aspects of park and ride facility?

DESCRIPTION	GOOD	VERYGOOD	BAD	VERY BAD
Availability of parking spaces				
Safety and security				
Cleanlinesss				
Ease of access				
Availability of public transport				
Cost of parking				
Overall experience				

15. How do you usually get from the park and ride facility to your final destination.

    a. Bus ( ).
    b. Train ( ).
    c. Carpool ( ).
    d. Bicycle ( ).
    e. Walking ( ).
    f. Others ( )

16. What improvements would you like to see at the park and ride facility?

a. More parking spaces ( )

b. Improved lighting and security ( )

c. Better signage and directions ( )

d. More frequent public transport services ( )

e. Enhanced amenities rest rooms ( ), waiting areas ( ), low parking fees ( )

f. Electric vehicle charging facilities ( )

g. Others ( )

17. Do you have additional comments or suggestion for improving the park and ride facility?